

Barra China A Total Market Equity Model for Long Term Investors Descriptor Details

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The 16 style factors of Barra China A Total Market Equity Model for Long Term Investors comprise a total of 40 descriptors. This document defines these descriptors in the style factors. The descriptors are listed under the style factors to which they belong.

Style: **Beta**

Descriptors: HBETA

Historical Beta

Computed as the slope coefficient from a time-series regression of stock excess returns against the cap-weighted excess returns of the estimation universe over a trailing window of 504 trading days, with a 252-day half-life.

The returns are aggregated over four-day windows to reduce the effect of non-synchronicity and auto-correlation.

Style: **Book-to-Price**

Descriptors: BTOP

Book-to-Price

Computed by dividing the last reported book value of common equity by the current market capitalization.

Style: **Dividend Yield**

Descriptors: DTOP

Dividend-to-Price

Computed by dividing the trailing 12-month dividend per share by the price at the last month end.

DTOPF

Analyst-Predicted Dividend-to-Price

Computed by dividing the 12-month forward-looking dividend per share (DPS) by the current price.

Style: **Earnings Quality**

Descriptors: ABS

Accruals - Balance Sheet Version

The balance-sheet-based accruals, $ACCR_BS$, is first computed from (i) consecutive changes in balance sheet items and (ii) depreciation. The descriptor is then computed as the negative $ACCR_BS$ normalized by total assets:

$$ABS = -ACCR_BS/TA$$

Thus, it goes long companies with low accruals.

ACF

Accruals - Cash flow Statement Version

The cash-flow-statement-based operating accruals, $ACCR_CF$, is first computed from (i) items from the most recent cash flow statement and (ii) depreciation. The descriptor is then computed as the negative $ACCR_CF$ normalized by total assets:

$$ACF = -ACCR_CF/TA$$

Thus, it goes long companies with low accruals.

Style: **Earnings Variability**

Descriptors: VSAL

Variability in Sales

Computed by dividing the standard deviation of the annual sales of the last five fiscal years by the average annual sales.

VERN

Variability in Earnings

Computed by dividing the standard deviation of the annual earnings of the last five fiscal years by the average annual earnings.

VFLO

Variability in Cash-flows

Computed by dividing the standard deviation of the annual cash flows of the last five fiscal years by the average annual cash flow.

ETOPF_STD

Standard deviation of Analyst Forecast Earnings-to-Price

Computed by dividing the standard deviation of the 12-month forward-looking earnings per share estimates by the current price.

Style: **Earnings Yield**

Descriptors: CETOP

Cash-Earnings-to-Price

		Computed by dividing the trailing 12-month cash earnings by the current market capitalization.
	ETOP	<u>Earnings-to-Price</u>
		Computed by dividing the trailing 12-month earnings by the current market capitalization.
	EM	<u>Enterprise Multiple (EBIT to EV)</u>
		Computed by dividing the earnings before interest and taxes (EBIT) from the last fiscal year by the current enterprise value (EV).
	ETOPF	<u>Analyst-Predicted Earnings-to-Price</u>
		Computed by dividing the 12-month forward-looking earnings by the current market capitalization.
Style:	Growth	
Descriptors:	EGRLF	<u>Analyst Predicted Earnings Long-term Growth</u>
		Long-term (3-5 years) earnings growth forecasted by analysts.
	EGRO	<u>Earnings per Share Growth Rate</u>
		Computed by dividing the slope coefficient from the regression of the annual earnings per share from the last five fiscal years against time, by the average annual earnings per share.
	SGRO	<u>Sales per Share Growth Rate</u>
		Computed by dividing the slope coefficient from the regression of the annual sales per share from the last five fiscal years against time, by the average annual sales per share.
Style:	Investment Quality	
Descriptors	AGRO	<u>Total Assets Growth Rate</u>
		Computed by first dividing the slope coefficient from the regression of the total assets from the last five fiscal years against time by the average total assets, and then multiplying by -1 to reverse the sign.
	IGRO	<u>Issuance Growth</u>
		Computed by first dividing the slope coefficient from the regression of the number of shares outstanding from the last five fiscal years against time by the average number of shares outstanding, and then

multiplying by -1 to reverse the sign.

CXGRO

Capital Expenditure Growth

Computed by first dividing the slope coefficient from the regression of the capital expenditures from the last five fiscal years against time by the average capital expenditures, and then multiplying by -1 to reverse the sign.

Style: **Leverage**

Descriptors: MLEV

Market Leverage

Computed as,

$$MLEV = \frac{ME + PE + LD}{ME}$$

where ME is the market value of common equity on the last trading day, and PE and LD are the preferred equity and long-term debt, respectively, from the last fiscal year.

BLEV

Book Leverage

Computed as,

$$BLEV = \frac{BE + PE + LD}{BE}$$

where BE, PE, and LD are the book value of common equity, preferred equity, and long-term debt, respectively, from the last fiscal year.

DTOA

Debt-to-Assets

Computed as,

$$DTOA = \frac{TL}{TA}$$

where TL and TA are the total liabilities and total assets, respectively, from the last fiscal year.

Style: **Liquidity**

Descriptors: STOM

Monthly Share Turnover

Computed as the log of the percentage of shares traded in the most recent month.

STOQ

Quarterly Share Turnover

Computed as the log of the average percentage of shares traded

monthly over the last three months.

STOA

Annual Share Turnover

Computed as the log of the average percentage of shares traded monthly over the last 12 months.

ATVR

Annualized Traded Value Ratio

Computed as the exponentially-weighted sum of the percentage of shares traded daily over a trailing 252-day window with a 63-day half-life.

Style: **Long-Term Reversal**

Descriptors: LTRSTR

Long-term Relative Strength

The non-lagged Long-term Relative Strength is first computed as the exponentially-weighted sum of the log excess returns of the stock relative to the market over a trailing 1040-day window, with a 260-day half-life. The final LTRSTR descriptor is then computed by first taking the equal-weighted average of the non-lagged values over an 11-day window lagged by 273 days, and then multiplying it by -1 to reverse the sign.

LTHALPHA

Long-term Historical Alpha

The non-lagged Long-term Historical Alpha is first computed as the intercept term from a CAPM regression similar to the one used to compute the HBETA descriptor, except with a 1040-day window and a 260-day half-life. The final LTHALPHA descriptor is then computed by first taking the equal-weighted average of the non-lagged values over an 11-day window lagged by 273 days, and then multiplying it by -1 to reverse the sign.

Style: **Mid Capitalization**

Descriptors: MIDCAP

Cube of Size Exposure

The Size factor exposure is first cubed, and then orthogonalized to Size on a regression-weighted basis, and finally winsorized and standardized.

Style: **Momentum**

Descriptors: RSTR

Relative Strength 12-month

The non-lagged Relative Strength is first computed as the exponentially-weighted sum of the log excess returns of the

stock relative to the market over a trailing 252-day window, with a 126-day half-life. The final RSTR descriptor is then computed as the equal-weighted average of the RS over an 11-day window lagged by 11 days.

HALPHA

Historical Alpha

The non-lagged Historical Alpha is first computed as the intercept term from the same time-series regression that is used to compute HBETA. The final HALPHA descriptor is then computed as the equal-weighted average of the non-lagged values over an 11-day window lagged by 11 days.

Style: **Profitability**

Descriptors: ATO

Asset Turnover

Computed as,

$$ATO = \frac{Sales}{TA}$$

where Sales is the trailing 12-month sales, and TA is the most recently reported total assets.

GP

Gross Profitability

Computed as,

$$GP = \frac{Sales - COGS}{TA}$$

where Sales, COGS, and TA are the sales, cost of goods sold, and total assets, respectively, from the last fiscal year.

GPM

Gross Profit Margin

Computed as,

$$GPM = \frac{Sales - COGS}{Sales}$$

where Sales and COGS are the sales and cost of goods sold, respectively, from the last fiscal year.

ROA

Return on Assets

Computed as,

$$ROA = \frac{Earnings}{TA}$$

where Earnings is the trailing 12-month earnings, and TA is the most recently reported total assets.

Style: **Residual Volatility**

Descriptors: HSIGMA

Historical Sigma

Computed as the volatility of the residual returns from the same time-series regression that is used to compute HBETA.

DASTD

Daily Standard Deviation

Computed as the volatility of daily excess returns over the past 252 trading days with a 42-day half-life.

CMRA

Cumulative Range

Computed as the gap between the highest and lowest points of the cumulative log excess return in the past 12 months.

Style: **Size**

Descriptors: LNCAP

Log of Market Capitalization

Computed as the natural logarithm of the market capitalization of the firm.

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